

A Chatroom, Independent Attempts, and One Agent on Three Tasks

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Abstract

We compare a single agent, a three-seat chatroom, and selection from independent attempts on machine-scored tasks at roughly matched mean spend. All results are dated measurements of `claude-sonnet-5` under the `sonnet` alias on recorded subscription accounts, not estimates for other models or deployments. In Study 1, the chatroom passed both interpreter tasks on every seed, against 61/80 single-agent passes across their shared task family. The vote arm passed 75/80; direct tests did not distinguish it from the chatroom. On a numeric-formatting task, one agent passed 14/40, the chatroom 6/40, and the vote 7/40. The lower chatroom rate was significant under the pre-registered test but not a later Fisher test. Voting selected the common wrong implementation even though correct attempts were often available. The vote prediction succeeded on one interpreter task but failed its stated significance clause on the other.

The same recorded alias and settings yielded sharply different thinking-token distributions across subscription accounts. Study 2 mostly ran on the long-thinking account, leaving its pre-registered short-thinking primary comparisons essentially unobserved. Its small, exploratory long-thinking comparisons distinguished no arm from one agent. The comparison shows why serving-account provenance belongs with each run’s outcome.

1 Introduction

A chatroom can give several language-model agents a chance to challenge and revise one another’s work. It also buys more attempts at the task. To assess what the conversation contributes, the relevant comparison is with independent attempts chosen without seeing the answer, at a similar cost. This paper makes that comparison on a small, deliberately difficult task suite, then shows how a change in the serving account limited a later, broader study.

The measured system is `agent-chatroom-mcp`. Every run reported here used `claude-sonnet-5` under the `sonnet` alias between 18 and 20 September 2026. The subscription account active at run time is reconstructed from a sanitized switch log. Pass rates, costs, and which task was near ceiling belong to those dates, model and accounts. They are not predictions for newer models. The study method and the need to retain serving provenance do carry forward; the observed minority-versus-majority pattern is a hypothesis to test elsewhere.

Study 1 used seeds 101–140 on three fixed tasks, comparing one agent (A), a three-seat chatroom (C), and independent attempts with oracle-free selection (K). The third task asks for exact numeric formatting. All outcomes came from a machine oracle, not from the agents’ own verdicts.

Study 2 used seeds 501–520 and added a higher-effort single agent (AH) and a builder-reviewer pair (B). Its pre-registered primary comparison targeted short thinking. The run stopped with missing cells after most work was served by the long-thinking account. The resulting long-thinking comparisons were specified after data had been seen and remain exploratory.

The paper makes three narrow points. In Study 1, direct tests did not distinguish the chatroom from independent selection, even though both could outperform one agent on the interpreter family and both had lower observed success on `printf`. Across recorded accounts, thinking tokens differed under the same alias and settings, with one long-thinking single agent near ceiling on the measured interpreter task. Study 2 cannot answer its primary question: the intended short-thinking stratum was almost absent.

2 Evaluated system

The evaluated build of `agent-chatroom-mcp` gave each chatroom seat an independent opening before shared discussion. Seats could propose and amend one answer, challenge a quoted span, and vote on the current proposal. Agreement votes had to quote the proposal; the room used a supermajority rule that required unanimity with three active seats. A verification gate required a machine-parseable verdict from a connection other than the proposer, and the hub assigned that reviewer. The verify gate and reviewer assignment were present before both studies. Kick, replace and heartbeat changes dated 22 September were later additions and were not evaluated here. Study 1 chatroom cells record hub entry hash `66f9d135...`; run artifacts carry the revision used for each cell.

Arm A used one agent on the same public task brief. Arm K ran fresh, non-communicating A-style attempts and selected one answer without access to the hidden oracle. For the two interpreter tasks, it chose the most frequent normalized final answer, breaking a tie by sub-seed. For `printf`, it ran candidate programs on a fixed public-specification probe set and chose the implementation with the greatest execution agreement. Arm B in Study 2 passed one builder’s output to a reviewer for possible revision; AH used one agent at higher effort. Thus B was run, while Study 1 has no B cell.

3 Method

Tasks and scoring. Study 1’s tasks were admitted after a single-agent pilot screened about 45 designs for a discriminating pass rate. The admitted tasks each contain one known trap. `stamp-interpreter` and `stamp-2` ask for programs implementing the same invented-language scoping rule; they are two fixed instances of one family, not independent evidence from two domains. `bench-printf-format` asks for exact formatting where a natural implementation using the host runtime’s shortest representation disagrees with the specified output. The same hidden, machine-checked fixtures scored every arm. The seeds vary attempts on fixed task instances, rather than sampling new tasks. The task suite therefore supports comparisons on these traps, not an estimate across software engineering work.

The pilot targeted single-agent pass rates between 0.3 and 0.7, inclusive. The interpreter admission pilots used ten seeds each; `printf`’s retained raw-backed pilot used fifteen. Choosing `printf v1` over `v2` was a post-data selection, as recorded in `paper/amendments.md`.

Study 1 design. For each task, A and C ran on seeds 101–140, interleaved per seed. A’s budget ceiling was tied to C’s realized spend; matching did not force the same realized spend. K was separately pre-registered after A and C results but before K ran, on the same seeds and tasks. Its number of attempts was chosen from the earlier mean A and C costs, with each attempt run afresh. The selector had no oracle input. The fixed K group sizes were ten attempts for `stamp-interpreter`, eight for `stamp-2`, and seven for `printf`. The per-attempt cap left room for an ordinary A-style

attempt; total group cost included every attempt even if it was killed or not selected. The matching target was mean realized C spend, not identical spend on every seed.

K ran from 13:22 to 15:33 UTC, before the account switch recorded at 16:38 UTC; an earlier shift-time estimate was superseded by that switch log. The comparison between K and C still spans separate collection windows. K exceeded its paired C spend on 19, 10, and 9 of the 40 seeds for the three tasks, despite matching at the mean.

The original A–C protocol specified a two-proportion z-test, using Fisher’s exact test for small cells, separately by task. After the data, the two interpreter tasks were pooled as a family and A–C was retested with Fisher’s exact test and Holm correction across the two families. We report that revised family result alongside the original per-task tests. The K pre-registration fixed six two-sided Fisher comparisons, K against A and C on each task, with Holm correction across all six. Pooled K comparisons on the interpreter family are exploratory and do not replace those registered tests.

The common seed labels put arms on the same task brief, but the attempts are separate model runs; the Fisher calculations do not treat them as paired observations.

Study 2 design and missingness. Study 2 specified five arms on seeds 501–520: A, AH, B, K and C. Its primary tests used the short-thinking stratum. The account switcher was not known to the harness at launch. A sanitized log later associated each run with its active account; mixed-account seeds were separated from within-regime comparisons. The run stopped at 190/200 cells, leaving ten absent. Printf seed 501 is kept in descriptive totals but excluded from inference after a documented cap deviation. One printf chatroom run has unknown cost. The long-thinking tests were added after much of that stratum had been observed, and use 19 seeds per arm on `stamp-interpreter` and 15 on printf. They use Fisher tests with Holm correction across the registered comparison family of fourteen comparisons, but their timing makes them exploratory. Strata condition on A’s realized post-run thinking, rather than an independent pre-run measurement; its sentinel also cannot rule out a change later in the same seed.

Account measurement. The account analysis joins single-seat run intervals to switch times and uses only runs with thinking telemetry for the thinking-token comparison. Long thinking was defined in the Study 2 pre-registration as at least 4,000 thinking tokens. Five of 1,947 eligible runs straddled a switch and were excluded. Output-token distributions give a second descriptive account comparison, but output across a three-seat room is not directly comparable to a single seat. Neither the active account nor effort was randomized. The client records the association, not the server-side reason for it.

4 Results

4.1 Study 1: conversation and independent selection

Table 1 gives the A–C family comparison; Figures 1 to 3 show the same Study 1 results as charts. Across the two interpreter programs, A passed 61/80 and C passed 80/80. The family-pooled Fisher test, adopted after the runs, gives $p = 0.000001$ before Holm adjustment. The original pre-registered per-task tests also rejected A–C on each interpreter program; the pooled result is a revised analysis, not the pre-registered test. K passed 36/40 on `stamp-interpreter` and 39/40 on `stamp-2`, or 75/80 pooled. Its pooled comparison with A ($p = 0.0033$) is exploratory. The registered K tests show a narrower result: K beat A on `stamp-2` (Holm $p = 0.009$), but was not

Table 1: Study 1, A versus C by task family.

Family	A pass/ <i>n</i>	C pass/ <i>n</i>	Fisher <i>p</i> (raw)	Holm <i>p</i> (<i>m</i> = 2)	Sig.
interpreter (stamp-interpreter + stamp-2)	61/80	80/80	0.000001	0.000002	yes
printf (bench-printf-format)	14/40	6/40	0.069165	0.069165	no

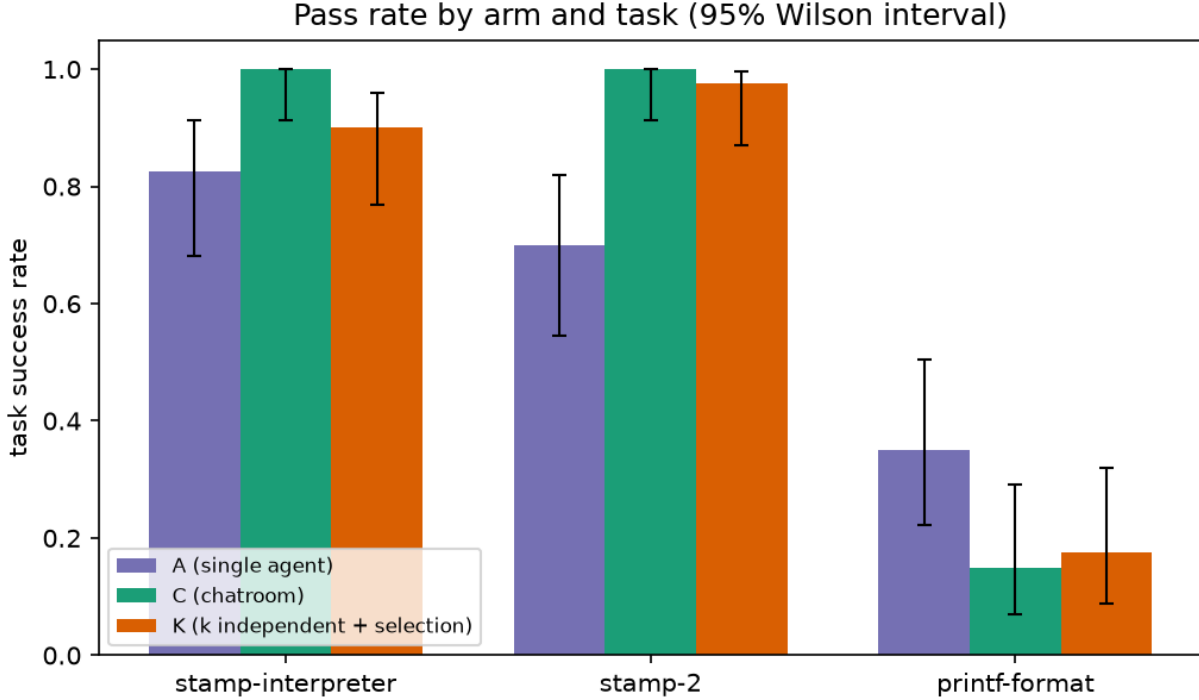


Figure 1: Study 1 task success by arm and task, with Wilson intervals at the 95 per cent level; 40 seeds per cell, short-thinking account, 19 September 2026. The chatroom (C) passed 40/40 on both interpreter tasks; on printf both aggregation arms fell below one agent (A).

distinguished from A on **stamp-interpreter** (Holm $p = 1.000$). The pre-registered prediction said K would beat A on both. That clause failed.

Direct K–C tests did not distinguish the arms on any Study 1 task: every Holm-adjusted p was at least 0.578. Pooling the interpreter tasks gives 75/80 for K and 80/80 for C, with an exploratory Fisher $p = 0.059$. Neither result establishes equality. In particular, a significant C–A result and a nonsignificant K–A result on **stamp-interpreter** are not themselves a test of C against K. The direct comparison is the relevant test, and the available sample leaves a small difference unresolved.

On **bench-printf-format**, A passed 14/40, C 6/40, and K 7/40. The pre-registered A–C z-test gave $p = 0.039$; the later family-based Fisher test gave $p = 0.069$, so the significance decision depends on the stated test. K was not distinguished from A (Holm $p = 0.578$) or C. The observed rates on this task are lower for both aggregation arms; they do not support a general assertion that either method lowers success on other tasks.

The independent-attempt evidence explains K’s printf outcome more specifically. In a fixed pool of 40 A attempts, all 14 passing implementations agreed on every one of 9,650 public-specification probes, and all 26 failing implementations also agreed with one another. Thus execution agreement

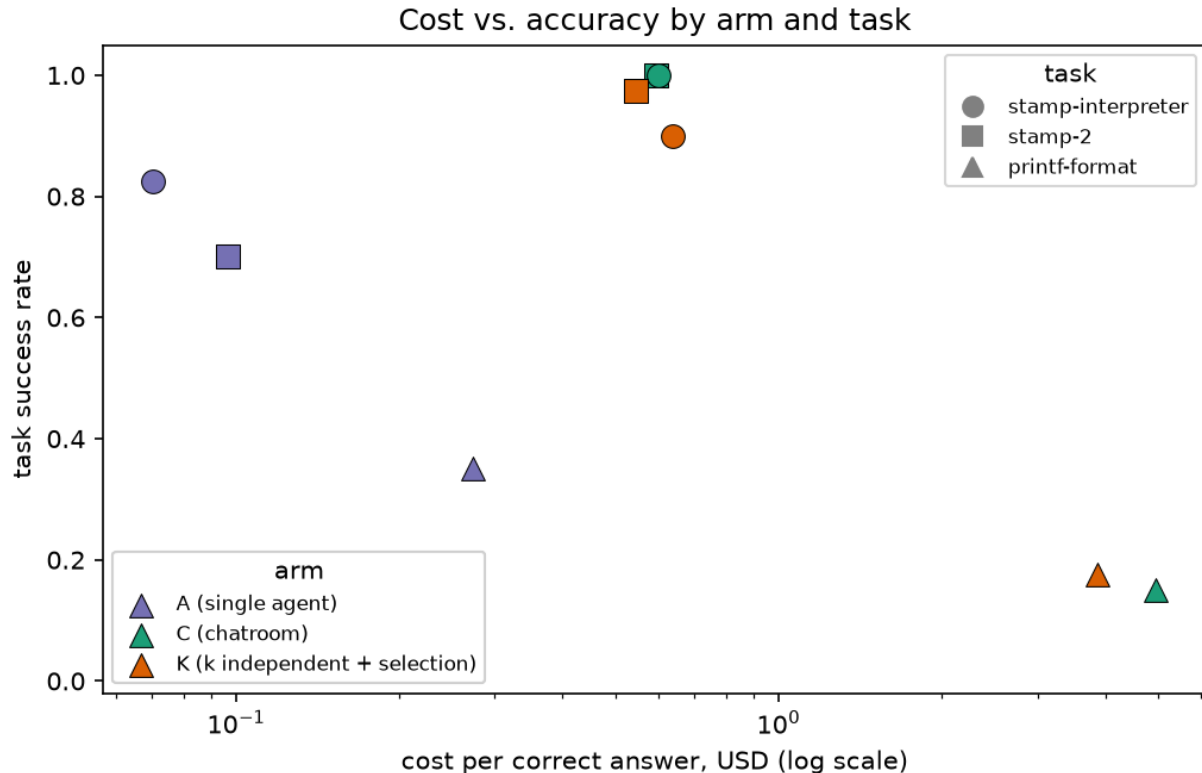


Figure 2: Study 1 cost per correct answer (log scale, API list price) against success rate, one point per task and arm. One agent was the cheapest per correct answer on every task; the chatroom and the vote cost about the same as each other on the interpreter tasks, and the vote cost less on printf (\$3.87 against \$4.97).

identified the larger wrong cluster. In the real K groups, at least one correct attempt was available in 37/40, but the oracle-free selector submitted a correct one in only 7/40. The availability count is an oracle ceiling, not a deployable arm. The C artifacts retain pass or fail, but not the failing implementations; the same modal-error explanation for C is an association, not an observed C selection mechanism.

The interpreter answers show the opposite population pattern in Study 1. For **stamp-interpreter**, the seven A failures among 40 were identical; for **stamp-2**, 11 of the 12 failures were identical and one differed. Here the correct answer was the majority of A answers in the observed pool. The room summaries add diagnostic context: at least 26 of 40 **stamp-interpreter** rooms and 14 of 40 **stamp-2** rooms reported an erroneous opening followed by another seat’s correction. These are manual lower bounds from final seat summaries, without retained raw room events. They do not identify a causal chatroom mechanic.

The extra computation had a substantial price in these runs. C’s cost per correct answer relative to A was 8.56 times on **stamp-interpreter**, 6.16 times on **stamp-2**, and 18.17 times on printf. K and C cost respectively \$0.637 and \$0.602 per correct answer on **stamp-interpreter**, \$0.545 and \$0.595 on **stamp-2**, and \$3.87 and \$4.97 on printf. These are list-price calculations under the recorded accounts and include all K attempts, including those not selected.

Arm K mechanism: does agreement track correctness?

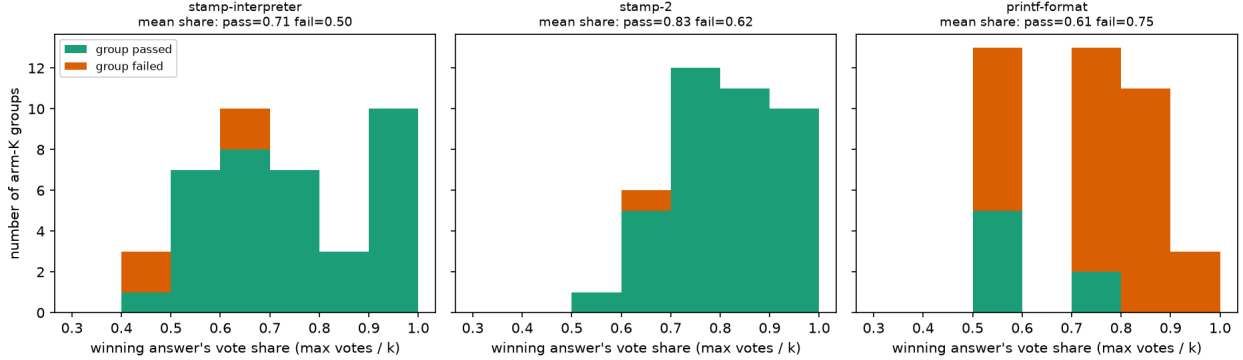


Figure 3: Study 1 vote arm: in each group, the winning answer’s share of the votes, split by whether the selected attempt passed. On printf, strong agreement goes with failure, because the failing implementation was the larger cluster of independent attempts.

Table 2: Recorded subscription account and run-level thinking or output tokens. Generated from the account-switch log and committed run artifacts.

Active account	Runs	Long-thinking runs	Median thinking tokens	Min	Max
Account 1	40	0/40	968	463	1295
Account 3	11	0/11	716	595	1316
Account 4	385	384/385	7709	3742	31604

Task family	Active account	Runs	Median output tokens	Min	Max
bench-printf-format	Account 1	337	4645	3789	6595
bench-printf-format	Account 3	26	4225	3726	5181
bench-printf-format	Account 4	152	25144	16212	43700
stamp-interpreter	Account 1	590	1716	1357	2232
stamp-interpreter	Account 4	313	7310	4494	16896

Single-seat runs (arm A, arm AH, and the constituent attempts of arm K) recorded in `bench/results`, joined by start time to a sanitized extract of the maintainer’s account-switcher log (`bench/results/account-switch-log/switches.json`: switch times and account slot numbers only). 5 of 1947 runs straddle a switch and are left out. Top: runs that recorded thinking tokens; long thinking means at least 4000 thinking tokens, the threshold frozen in `paper/prereg-confirmatory.md`. Bottom: output tokens, which every run recorded, for the two Study 2 task families. Account numbers are slots in the switcher, not identifiers.

4.2 Account-associated thinking

Table 2 joins single-seat run times to the sanitized switch log. Among runs with thinking telemetry, account 4 had 384/385 at or above 4,000 thinking tokens. Accounts 1 and 3 had none among 51 combined runs. Median thinking was 7,709 tokens on account 4, against 968 and 716 on accounts 1 and 3. Within each of the two task families shown, account 4’s output-token range lay entirely above those of the other measured accounts. These are strong associations under the same alias and settings, not evidence that the client can identify what the provider changed.

The association matters for interpreting accuracy. Short-thinking single-agent measurements on `stamp-interpreter` came from several cohorts: the admission pilot, the Study 1 A grid, constituent K attempts, a cap control, and the window control. Their estimates ranged from 0.475 to 0.825. The Study 1 grid’s 33/40 was therefore one estimate within an observed span, not a stable model parameter. The separate long-thinking window-control cohort had 78/80 successful constituent single attempts. Study 2’s long-thinking A arm passed 18/19 on that task and cost \$0.136 per

Table 3: Study 2 descriptive outcomes by task, stratum and arm. Inference excludes mixed-account seeds and printf seed 501; the eligible long-thinking comparisons use fewer seeds than some descriptive rows.

Descriptive rows retain printf seed 501; it is excluded from both primary and exploratory comparisons.						
Family	Stratum	A	AH	B	K	C
stamp-interpreter	long thinking	18/19	19/19	19/19	19/19	19/19
stamp-interpreter	mixed accounts	1/1	1/1	1/1	1/1	1/1
bench-printf-format	short thinking	0/1	0/1	1/1	0/1	0/1
bench-printf-format	long thinking	12/16	12/16	11/16	13/16	10/16
bench-printf-format	mixed accounts	0/1	0/1	1/1	1/1	0/1

correct answer. These are different cohorts, not a randomized account comparison; their gap does not show that either account or effort caused an accuracy change. The source of the variation among short-thinking estimates is also unidentified. **stamp-2** was not measured under long thinking, so the long-thinking interpreter result is confined to **stamp-interpreter**.

4.3 Study 2: the primary stratum was nearly absent

Study 2’s pre-registered primary used short-thinking seeds. It obtained no eligible short-thinking **stamp-interpreter** seed and only one printf seed. The interpreter’s apparent short-thinking seed crossed accounts between arms and was excluded from within-regime inference. One printf seed also crossed accounts. Table 3 retains those rows descriptively; the printf seed with a cap deviation is likewise descriptive only. The missing ten cells and excluded seeds leave the registered primary comparisons essentially unobserved.

Unlaunched cells are absent from the denominators; launched cells with non-pass outcomes count as failures. Unknown cost is not treated as zero.

Within the exploratory long-thinking stratum, the interpreter comparisons use 19 seeds per arm. A passed 18/19; AH, B, K and C each passed 19/19. On printf, the inferential sample has 15 seeds per arm: A and AH passed 11/15, B and C 10/15, and K 12/15. Every Holm-adjusted comparison against A had $p = 1.000$. That test result means no arm was distinguished in these small samples, not that their effects are equal or that the primary prediction was confirmed. On the interpreter, B cost \$0.269 per correct answer and C \$1.407, with both passing 19/19. The pair’s observed cost was about a fifth of the chatroom’s. The pre-registered cost prediction concerned short thinking; this exploratory long-thinking inequality does not confirm it.

5 What this does not show

The planned vote-versus-chatroom comparison under one stable serving condition was never run at adequate power.

The maintainer’s working pipeline—prompt, prior-art agent, plan, build, and devkit review—has never been compared with one agent, a builder-reviewer pair, or a room. The marginal value of review by a non-author was not identified either. B ran in Study 2 and in a small pilot, but the Study 2 primary was essentially absent. A separate build-suite diagnostic reached a 9/9 ceiling in every cell.

The bundled chatroom build was compared with other arms, but its individual mechanics were not ablated. Regression tests show that gates and assignments operate as coded; one bundled-build trial is not a causal estimate for a gate. No effect size is assigned here to blind openings, challenge, amended proposals, quote-checked votes, reviewer assignment, or verification.

Other engineering observations are outside the task-success comparison. Held waits were measured at 115.6 million versus 53.6 million tokens in one room each on different tasks, so that contrast is observational. OpenRouter sticky-session caching was mock-tested only. Checkpoint trimming has no answer-quality test. None of these results supports a claim about the measured A, B, C, or K accuracy differences.

6 Related work

Multi-agent debate provides the immediate setting for this comparison. Du et al. [1] study debate as a route to better answers, while Smit et al. [5] examine how debate strategies compare with simpler baselines. Those papers motivate asking whether a conversational protocol earns its additional cost. Our measured question is narrower: on three fixed machine-scored tasks, did this recorded chatroom differ from a vote over fresh attempts?

Arm K’s interpreter selector is a form of self-consistency: independently generated answers are normalized and the most frequent is submitted [7]. Its code selector instead uses execution agreement among candidate programs, following the type of signal studied by Shi et al. [4]. In the observed printf pool, agreement favored the common wrong implementation. That is why the candidate-availability ceiling and the submitted-answer rate must be reported separately. The result describes the selector as used here; it is not a comparison among alternative selectors.

The chatroom’s independent openings, challenge, proposal and verification steps make it tempting to attribute corrected answers to a particular interaction. Work on divergent thinking and premature consensus [3; 8] gives reasons to inspect those interactions. Work on minority answers in multi-agent debate [2] likewise frames the wrong-cluster observation as a testable question rather than an established general rule.

Finally, the hidden machine oracle keeps the agents’ own consensus separate from correctness. Research on language-model judges [9; 6] concerns a different measurement route; none of the pass rates here uses such a judge. Our main measurement complication arose earlier, at model serving: the same recorded alias and settings accompanied different thinking-token distributions on different active accounts. The account log and token telemetry therefore belong alongside accuracy and cost in any attempt to reproduce these particular comparisons.

A Reproducibility

The repository’s `paper/README.md` gives the commands for inspecting the five-arm snapshot, regenerating tables and figures, auditing prose numbers, and building the manuscript. The two study specifications are `paper/protocol.md`, `paper/prereg-arm-k.md`, and `paper/prereg-confirmatory.md`; dated deviations and exclusions are in `paper/amendments.md`. Those records distinguish a planned comparison from a later analysis.

The outcome artifacts are under `bench/results/rq1-suite`, `bench/results/rq1-arm-k`, and `bench/results/rq1-confirmatory`. The sanitized account switch times are in `bench/results/account-switch-log/switches.json`. Generated reports in `paper/generated` retain the per-arm denominators, prices, strata, and warnings used here. For the room-correction

tallies, `docs/arm-c-process-trace.md` states the manual criteria and names representative artifacts.

The manuscript inputs generated tables without editing them. `paper/regen-manifest.sh` names their generators and committed inputs. `scripts/paper-number-audit.ts` checks whether extracted prose decimals, fractions, and percentages have numeric matches somewhere in its source list, which includes the manual process-trace note. It does not check bare integers or establish that a matched number supports the surrounding claim.

Author and AI disclosure

Author. [Maintainer name, affiliation, and contact to be filled before submission.]

AI disclosure. Language-model agents helped build the evaluated system, run its recorded experiments, and draft this manuscript. The human author is responsible for checking the evidence, citations, and final submitted text. No agent is listed as an author.

References

- [1] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate, 2023. URL <https://arxiv.org/abs/2305.14325>.
- [2] Chuan He, Zebin Chen, Zhengyi Yang, Shaobo Qiao, Mingchen Ju, Jiate Liu, Dong Wen, and Guanfeng Liu. Minority sentinel: When to overturn majority voting in multi-agent llm debates, 2026. URL <https://arxiv.org/abs/2606.29270>.
- [3] Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Shuming Shi, and Zhaopeng Tu. Encouraging divergent thinking in large language models through multi-agent debate, 2023. URL <https://arxiv.org/abs/2305.19118>.
- [4] Freda Shi, Daniel Fried, Marjan Ghazvininejad, Luke Zettlemoyer, and Sida I. Wang. Natural language to code translation with execution, 2022. URL <https://arxiv.org/abs/2204.11454>.
- [5] Andries Smit, Paul Duckworth, Nathan Grinsztajn, Thomas D. Barrett, and Arnu Pretorius. Should we be going mad? a look at multi-agent debate strategies for llms, 2023. URL <https://arxiv.org/abs/2311.17371>.
- [6] Minsoo Song, Chanwoo Kim, Sugyeong Eo, and Chanjun Park. Beyond consensus: Downward bias and role asymmetry in multi-agent llm judges for subjective evaluation, 2026. URL <https://arxiv.org/abs/2608.30373>.
- [7] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models, 2022. URL <https://arxiv.org/abs/2203.11171>.
- [8] Binwei Yao, Chao Shang, Wanyu Du, Jianfeng He, Ruixue Lian, Yi Zhang, Hang Su, Sandesh Swamy, and Yanjun Qi. Peacemaker or troublemaker: How sycophancy shapes multi-agent debate, 2025. URL <https://arxiv.org/abs/2509.23055>.
- [9] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica.

Judging llm-as-a-judge with mt-bench and chatbot arena, 2023. URL <https://arxiv.org/abs/2306.05685>.